Telnet Configuration and Remote Access

1. Aim

To configure **Telnet** on a Cisco router in Cisco Packet Tracer and remotely access the router from a PC.

2. Introduction

Telnet is a network protocol used to remotely access and manage a device through a command-line interface. Telnet uses **TCP port 23**.

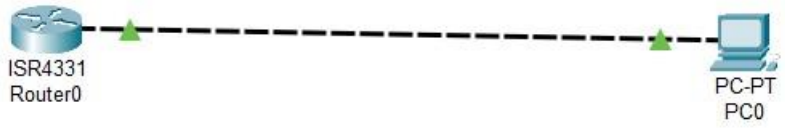
Unlike SSH, Telnet does **not encrypt** the communication. Therefore, usernames, passwords, and other data can potentially be viewed if the connection is intercepted.

In this practical, **PC0 acts as the Telnet client** and **Router0 acts as the Telnet server**.

3. Network Topology

The network contains:

- **1 Cisco ISR4331 Router (Router0)**
- **1 PC (PC0)**
- A network connection between the PC and router
- **Figure 1: Telnet Network Topology**



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· 4. IP Configuration

The PC and router were configured in the same network.

Device	IP Address	Subnet Mask
Router0	192.168.1.1	255.255.255.0
PC0	192.168.1.2	255.255.255.0

· 5. Router Configuration

First, the router interface was configured:

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitEthernet 0/0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0 Router(config-
if)# no shutdown
```

A username and password were created:

```
Router(config)# username admin secret cisco123
```

Then the VTY lines were configured to allow Telnet:

```
Router(config)# line vty 0 4
Router(config-line)# password harshika
Router# wr
```

· 6. PC Configuration

PC0 was configured with:

```
IP Address: 192.168.1.2
Subnet Mask: 255.255.255.0 Default
Gateway: 192.168.1.1
```

· 7. Testing the Connection

First, connectivity was checked using:

```
PC> ping 192.168.1.1
```

A successful ping confirms that PC0 can communicate with Router0.

The Telnet connection was then started using:

```
PC> telnet 192.168.1.1
```

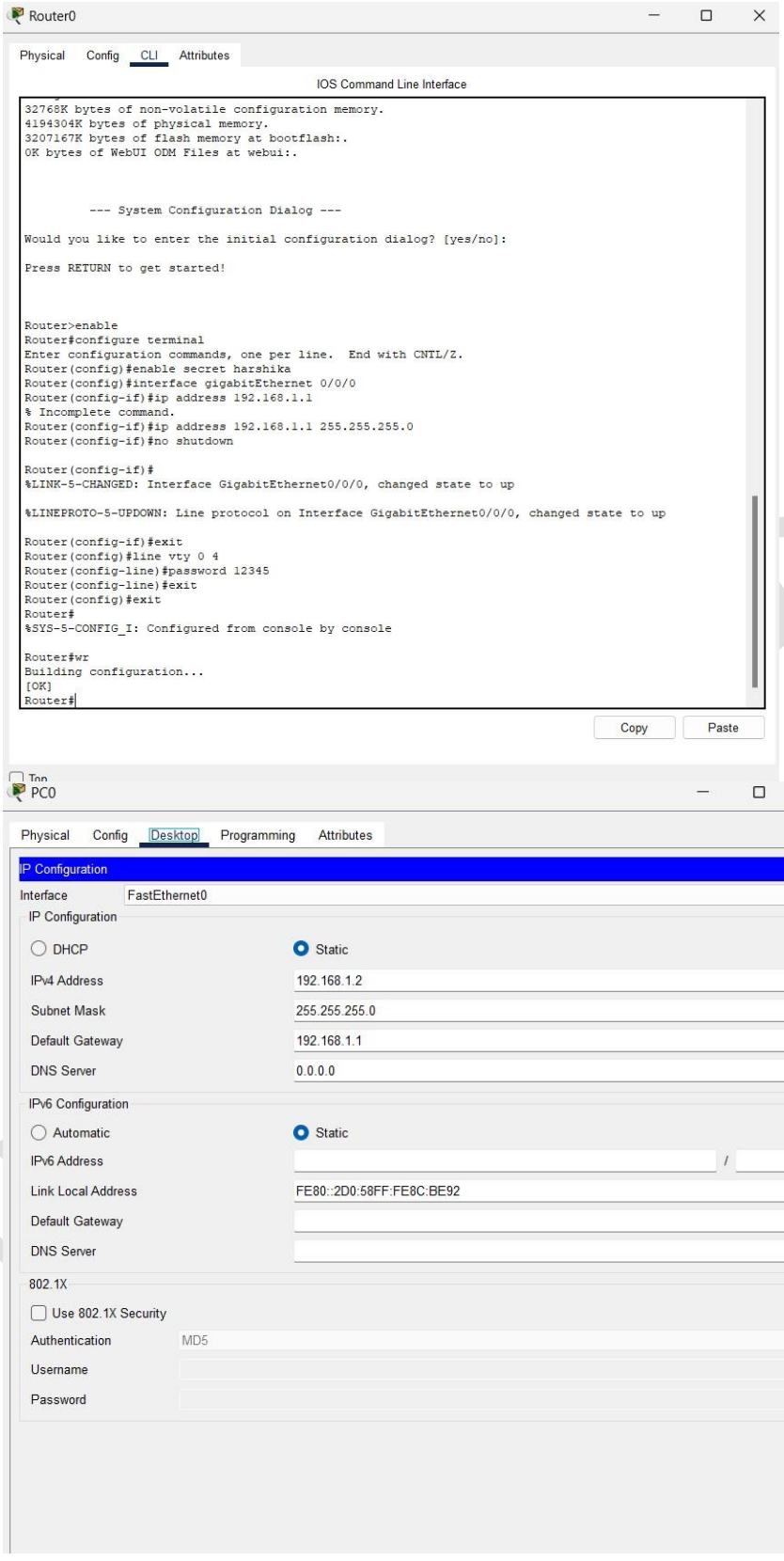
The router asks for the configured username and password.

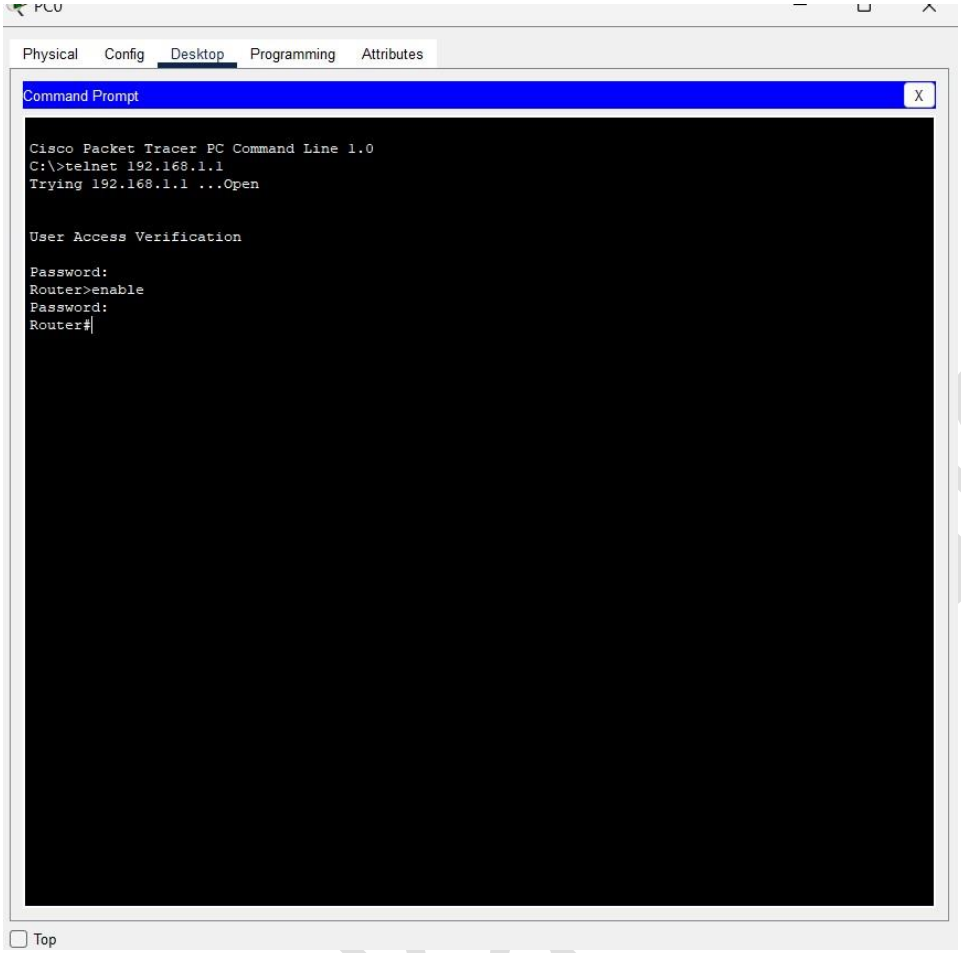
After successful authentication, the PC gets access to the router's command-line interface.

```
Router0>
```

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- **Figure 2: Successful Telnet Connection**





8. Result

The Telnet connection between **PC0 and Router0** was successfully established. PC0 was able to remotely access and manage Router0 using Telnet.

10. Conclusion

In this practical, Telnet was configured on a Cisco ISR4331 router using Cisco Packet Tracer. PC0 was used as the Telnet client and Router0 as the Telnet server. The connection was tested using ping and Telnet login. The practical demonstrated how Telnet provides remote command-line access to a network device.

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